

Understanding the inception of 14.7 Hz oscillations emerging from a data center

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ARTICLE INFO

Keywords:

Forced oscillations

Load dynamics

Spectral analysis

ABSTRACT

As Dominion Energy's power grid has grown to include an increasing number of inverter-based resources and power electronic components, we have begun to observe an increase in previously unforeseen and difficult-to-explain dynamic behaviors. In this paper, we present a measurement-based analysis of 14.7–14.8 Hz oscillations emerging from a data center in Dominion Energy's power grid. Synchrophasor data collected over several months were analyzed using non-parametric spectral analysis approaches to understand its inception and, more importantly, to characterize its behavior. What makes this case unique is the presence of an independent and unplanned exogenous excitation in the system. This excitation plays a critical role in understanding the inception and emergence of an otherwise unobservable dynamic process that, as it destabilizes, results in oscillations at a higher and narrow frequency range than that of conventional power system dynamics.

1. Introduction

1.1. Motivation

The growth of high-concentration large-scale commercial and industrial loads, including data centers, in Dominion Energy's system has increased on a considerable scale over the last decade. Furthermore, the concentration of such loads in populous regions, which is generally limited in local synchronous machine-based generation, poses challenges that should be addressed not only by the utility but also by large-scale commercial and industrial customers. Such challenges show the need for a coordinated and concerted effort to ensure that reliability is prioritized as more critical loads rely on it. Industrial loads such as data centers are not the only drivers, other inverter-based resources (IBRs, e. g. solar sites) and companies with investments in sensitive electronic manufacturing are just a few of those who could be impacted. The analysis of an oscillation emerging from a data center presented in this paper is a cautionary tale about how utilities and customers must work together to remain vigilant.

Challenges with data centers are especially of concern in areas that lack the tight interconnection of a 500KV system and instead operate in areas ripe for diluted reliability conditions at lower voltages, which can be exacerbated by weakened conditions such as shoulder-month

maintenance [1]. The interconnection of large-scale loads in relatively weak parts of the transmission system can cause local controllers to respond to dynamic performance irregularities. When controllers tuned for nominal operating conditions fail to account for significant changes on the grid, they can and have responded erratically. This has been observed in Dominion Energy's grid manifest as unwanted grid-to-data center controller interactions, as discussed in this paper, resonance across IBR controllers [2], and other behaviors [3]. A real-world example that motivates the work in this paper is presented in Fig. 2, which shows high-frequency undamped oscillations that appeared in a region rich in data centers in Dominion Energy's power system (a key use case of the work analyzed in this paper). Due to the nascent nature of observing such a phenomenon, operators and utilities alike are still learning what to analyze, how to monitor and how to optimally respond to such instances. In the case of this 'event', operators remained cautious and stoic but were unable to respond proactively. Fortunately, the oscillation in question lasted only two hours before organically disappearing, as it naturally damped from system changes.

1.2. Related works

Analyzing and modeling this behavior is difficult due to lackluster tools and visibility available to key decision makers. The inability to

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<https://doi.org/10.1016/j.segan.2025.101735>

Received 6 December 2024; Received in revised form 30 April 2025; Accepted 10 May 2025

Available online 12 May 2025

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access accurate transparent models involving customer power electronics means we, the utility, are forced to make educated guesses about what is behind the meter. For data centers, this manifests itself when Uninterruptible Power Supplies (UPS) provides frequency response services [4] that include complex power electronic controls [5] designed to protect the sensitive components housed within their server farms. However, we can only guess how such controllers are configured, leaving measurement data as the utility's only option for analysis and monitoring. In addition, the composition of a service region, i.e., the types of dynamic devices connected at a specific site and their operational schedule, may be completely unknown in worst-case scenarios; this is representative of the situation discussed in this paper.

It is worth noting that in the context of data centers, most of the literature is concerned with exploring how to meet the growing power demand for data centers from the point of view of planning and their operation. In the case of planning, data center power supply is being revisited due to regulatory and policy pressures, sustainability targets and ancillary service opportunities [6]. These catalysts are bringing operators of data centers to explore means to improve their energy efficiency [7], [8] and alternative back-up power sources to diesel-power systems, including renewable energy sources in the form of nearby distributed energy sources (DERs) [9] or from longer distances [10]. Moreover, a great effort has been made in finding ways to operate data centers in order to reduce their carbon footprint [11] and [12]. However, understanding the dynamic performance of data centers from real-world field measurements remains largely unaddressed in the literature, and this paper aims to bridge this gap.

To gain insight into the dynamic performance and operation of data centers, it is possible to leverage existing approaches used to understand power system dynamics, such as oscillation monitoring, that have proven successful in real-world applications. Detecting oscillations with low-damping, especially once they become unstable and cause sustained oscillations, is a well-studied area [13]. The techniques used to pinpoint 'events' seek large increases in measured spectral power across specific frequency bands – this allows for the detection of *known* phenomena. However, this approach poses a significant shortcoming, as by the time an oscillation is identified and detected, it may not be possible to mitigate it, resulting in damage or a decrease in quality of service. To address oscillations as the ones shown in Fig. 2, the system operators are left with only a few options and may feel compelled to take drastic measures, such as taking the offending asset offline or applying actions that may or may not resolve the issue. Meanwhile, addressing these issues post-facto, can result in necessary but high-cost solutions in the form of STATCOMS and SVC's [14]. Although they alleviate the symptoms and are essential to meet expected reliability standards, they do not address the underlying cause. Fortunately, in most cases of oscillations caused by unstable controllers, there are hints during the equipment's normal operation (ambient conditions), indicating that the controller is close to or on the verge of instability [15]. Therefore, identifying and promptly correcting emergent dynamic issues requires a shift in conventional engineering approaches, using for example the approach presented in this paper. The basis of our methodology is to screen through what has often been referred to as "noise," and use ambient data to intently identify unfavorable dynamics long before they occur [7].

The fundamental difficulty in working with ambient data is the stochastic nature of the system, which results from various uncertain and persistent inputs to the system [16]. In this regard, spectral analysis techniques [17] provide effective means to characterize the underlying system dynamics from ambient measurements (owing much to the nearly linear response of the system) and therefore have been widely used in mode estimation [16], [18], [19]. It's worth interjecting our concern here behind the broad assumption behind mode metering applications: while some oscillations (typically wide-area modes) operate under known frequency bands, this may not be the case for real-world control modes of converter-interfaced apparatuses [20,21], which makes the use of time-frequency analysis techniques necessary [22]. In

our experience, time-frequency analysis of the time window of the inception of converter-interfaced control modes yields unique insights into the oscillation mechanism, as will be demonstrated in this work.

1.3. Contributions

The primary contribution and novelty of this work lies in the application and synthesis of advanced measurement-based techniques to diagnose a complex, real-world, high-frequency (~ 14.7 Hz) oscillation event originating from critical power electronic loads (data centers). The root cause is attributed to the instability of a 10–11 Hz mode related to the data center's UPS system interacting with specific grid conditions, which manifests as the ~ 14.7 Hz oscillation (limit cycle) upon destabilization. Analyzing such phenomena using only field measurements, especially identifying the intricate inception mechanism involving mode transition and limit cycles triggered by exogenous inputs, represents a significant and timely contribution to the field, offering valuable insights for grid operators facing similar challenges. The goal of this work is not to share specific examples of observed phenomena, but rather to provide an evolving set of guidelines for performing data-driven analysis of local oscillations in situations where model scarcity makes such events difficult to diagnose.

1.4. Paper organization

The remainder of this paper is organized as follows; Section 2, gives necessary background on signal processing approaches utilized across this corpus of work and explains how they are used to gain insight from synchrophasor data. Section 3 provides an overview of a real-world oscillation event, as well as the region of interest. Section 4 performs analyses using the measurements. Finally, Section 5 summarizes the main conclusions of this work.

2. Time frequency analysis background

In this section we review key concepts used in the time-frequency analysis techniques that will be used to analyze measurement data.

2.1. Spectrum of power system ambient dynamics

For a stochastic process $x(t)$, power spectral density denoted by $S(f)$ is defined as the distribution of average signal power over frequency,

$$\lim_{T \rightarrow \infty} E \left(\frac{\int_0^T x^2(t) dt}{T} \right) = \int S(f) df \quad (1)$$

where, $E(\cdot)$ is the expectation operator. Processes with time-invariant mean and correlation functions are called wide-sense stationary and are characterized by a time invariant $S(f)$.

Although for most power system ambient dynamics (especially electromechanical modes) satisfy this assumption, it may not hold true in other dynamic processes contained in real-world data. The power system's time-varying nature results from

the changing operating conditions that are caused by changes in load, generation dispatch, etc., thereby making it a nonstationary process. Therefore, a more general characterization using a time-varying linear system driven by a stationary white noise input $u(t) \sim N(0, 1)$ is defined by

$$x(t) = \int h(t, \tau) u(t - \tau) d\tau = \int e^{j2\pi f t} H(t, f) dz(f) \quad (2)$$

where, $z(f)$ is a zero mean process with orthogonal increments, i.e., $E(dz(f)) = 0 \forall f$ and $E(dz(f) dz^*(f_1)) = S_u(f) \delta(f - f_1) df df_1 \forall (f, f_1)$ [23]. The PSD of $u(t)$ is denoted by $S_u(f)$ and is equal to a constant (equal to 1 here) for white noise. $h(t, \tau)$ represents the time varying impulse response and is defined as response at time t for impulse input at $t - \tau$,

$$H(t, f) = \int h(t, \tau) e^{-j2\pi f \tau} d\tau.$$

For nonstationary processes, a time varying notion for the spectrum, called the evolution spectrum (denoted by $S(t, f)$), was introduced in [23], for which the above process equals $|H(t, f)|^2$. Here, the most prominent dynamics of the underlying system appear as peaks (local maxima) in $S(t, f)$, which we are interested in capturing. Note that the discrete system changes, e.g. a change in network topology, are considered by assuming $S(t, f)$ to be piecewise continuous in t .

2.2. Time-frequency analysis using the wavelet transform

Due to the time-varying nature of real-world power systems [20], it is natural to utilize time-frequency analysis [10]. These techniques operate by decomposing the measurement signal into several constituent components, which (in general) correspond to distinct underlying phenomena in an actual system. In the linear class of techniques, the signal is characterized by inner products with a pre-assigned family of templates, which are normally skewed in the time and frequency plane.

In the wavelet transform [24], a pre-defined component called the ‘mother wavelet’ $\psi(t)$ is translated and scaled to create family of templates called ‘wavelets’. It is chosen in accordance with the dominant behavior in the measurements to be analyzed, thus providing great flexibility. Under the assumption that slow processes generally vary less frequently than fast ones, frequency resolution is increasingly favored over time resolution at lower frequencies. The Continuous Wavelet Transform (CWT) coefficient $W(a, b)$ for a signal $x(t)$ is obtained as,

$$W_x(a, b) = \int x(t) a^{-\frac{1}{2}} \bar{\psi}\left(\frac{t-b}{a}\right) dt \quad (3)$$

where, a is the scale factor, b is the shift factor and $\bar{\cdot}$ is the complex conjugate operator. Increasing a has the effect of stretching $\psi(t)$ (shift towards lower frequency), while b introduces a phase shift in time. In practice, the CWT is realized for sampled data by discretizing it as $a = 2^{\frac{j}{\nu}}$ for $j = 1, 2, 3, \dots$, ν is an integer > 1 called number of voices per octave and b takes integer values. Note that the CWT is a highly redundant transformation, that is, a signal x of N samples is converted to $M \times N$ wavelet coefficients with M being the number of scales. Note that this is needed for better characterization and localization, as illustrated later.

The mother wavelet $\psi(t)$ is designed to have compact time and frequency support. This means that translating and scaling is akin to moving around a band pass filter paired with a windowing function. In the present work, we use the Morlet wavelet, which consists of a complex sinusoid localized with a Gaussian window,

$$\psi(t) = e^{-\frac{2\pi^2 t^2}{k_0^2}} \times \left(e^{j2\pi t} - e^{-\frac{k_0^2}{2}} \right).$$

A major challenge in time-frequency analysis is to make a suitable trade-off between time and frequency resolutions, otherwise known as the Heisenberg’s uncertainty principle. Note that the CWT coefficients $W_x(a, b)$ are in a time-scale plane and not the time-frequency plane, where a single scale a is associated with a range of frequencies. To convert it to time-frequency representation, the synchrosqueezing wavelet transform was proposed in [25], which is a post-processing approach that allocates the wavelet coefficient $W_x(a, b)$ to points in time-frequency plane. This is achieved by estimating the dominant frequency $\omega_x(a, b)$ from $W_x(a, b)$ to which $W_x(a, b)$ value is allocated, resulting in

$$\omega_x(a, b) = -i(W_x(a, b))^{-1} \partial_b W_x(a, b). \quad (4)$$

3. Case study description

3.1. Study system

The region of Dominion Energy’s power grid under study is shown in Fig. 1 and is predominantly a 115 kV network rich in converter interfaced resources. A cluster of data centers at substations D1, D2, and D3 lies at the center of this region. Substations H and C have solar photovoltaic (PV) generation with 85 MW and 50 MW rating, respectively. A 267 MW hydro power plant is located at substation K, which serves as an important source of fault current. The nearest connection to higher kV levels is through a 115/230 kV transformer at substation C. This connects directly to a nearby 500/230 kV substation, which contains a ± 125 MVAR STATCOM.

3.2. Measurement data

The synchrophasor data used in this study is reported at 30 Hz by Digital Fault Recorders (DFRs). The DFRs perform 4800 Hz sampling of analog signals, which are then filtered and down sampled to 960 Hz from where synchrophasors at 60 Hz are estimated. The estimated phasors are down sampled to 30 Hz before reporting them to Dominion Energy’s Phasor Data Concentrator (PDC). Measurements are available at all the substations in Fig. 1 that are labelled with red letters, i.e., C, D1, D3, H and P. Note that there are no measurements at D2 and other unlabelled substations.

3.3. Event description

Personnel at substations B, C, and D2 noticed flickering during a summer day and reported it to operations, this was the first instance when the behaviour shown in Fig. 2 became of concern. Fig. 2 shows that the largest magnitude oscillation was observed in the data center region. The oscillations were preceded by four hydro units at K ramping off and being taken offline at 2:05 AM, labelled A in Fig. 2. At around 3:58 AM, labelled B in Fig. 2, the STATCOM setpoint was changed from 0 MVAR to -71 MVAR, that is, switching it to reactor operation, resulting in the oscillations vanishing shortly after.

To adequately determine the inception and emergence of the identified phenomena, a rigorous characterization of the oscillation was needed. This case also required to understand the role of all other dynamic devices and processes, to determine if they were involved and to analyse historical data that would help better explained the observed phenomena. In the next section, we attempt to address all these issues through a data-driven analysis, i.e., using measurements only.

4. Analysis

In the sequel, we present an analysis of the oscillatory phenomena presented above and verify it through historical data analysis of other similar instances. The analysis is divided in three parts, as shown in Fig. 3. First, the oscillation in Fig. 3 is characterized in the frequency domain to determine its frequency range. Once this has been determined, we analyse the mode shape of reactive and active power flows within the identified frequency range to make a more granular localization and to determine its governing control loop, i.e., whether it’s related to a controller that regulates active or reactive power. Second, we expand our analysis to understand the underlying mechanism that gives rise to its inception of the oscillation. This is achieved by exploiting the presence of periodic voltage sags in the signal, which are exploited as a form of exogenous excitation to the system serving to aid in the characterization of the inception mechanism observed. Lastly, the last part performs historical data analysis serving to better explain both the inception process and the emergence of the oscillation for multiple other instances of the phenomena analysed for a single instance. This last section serves both to verify and generalize our analysis of the original

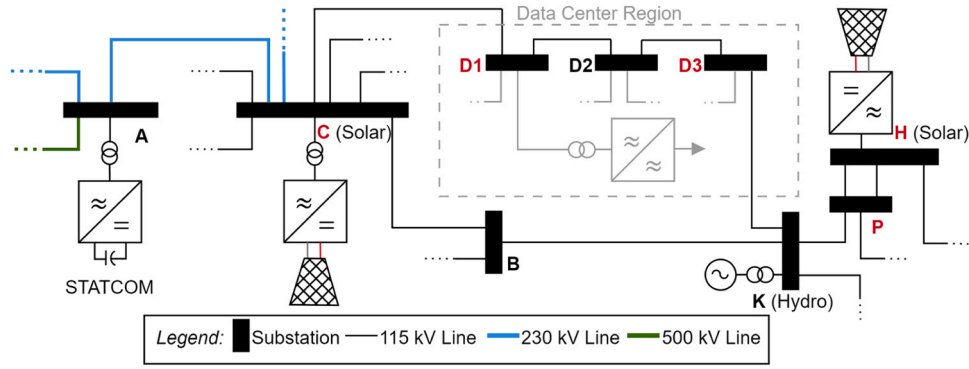


Fig. 1. Region of Dominion Energy's Power Network.

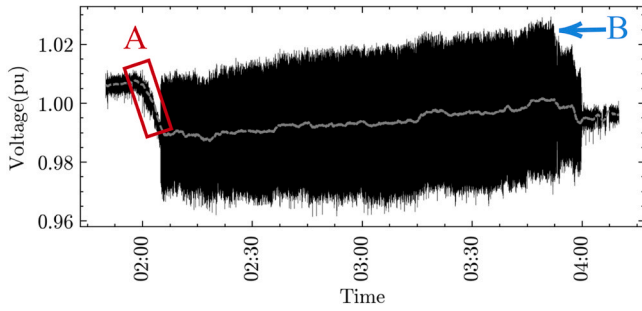


Fig. 2. Voltage Magnitude at Substation D3. Label A indicates when four hydro units were ramped down preceding the oscillation. Label B indicates when the STACOM setpoint was changed from 0 to -71 MVAR.

phenomena by considering multiple instances of it.

4.1. Oscillation characterization

4.1.1. Spectral analysis

The first step to identifying the potential source of an oscillation is to characterize the oscillation itself. To do so, we start by contrasting the measurement data pre-and-during the event by computing the voltage magnitude spectrum at substation D3, as shown in Fig. 4. Each graph was obtained by computing spectral estimates of 10 min windows, each using Welch's method [26] with a 2 min FFT window. The dark black line is the average across individual estimates.

Observing both Fig. 4a and b, it can be noted that there is a significant variability below 2 Hz range in both pre-and-during event time periods (labeled A), which indicates nonstationary behavior. Meanwhile, the behavior is largely stationary in roughly the 2–12 Hz range (labeled B). Furthermore, there is a family of spectral spikes at 1 Hz and their harmonics (indicated by a dashed square labeled C) in both pre-and-during event periods.

Next, in Fig. 4a's it can be observed that the PSD in the 10–12 Hz range most (labeled D) has a relatively higher value compared to that during the event. Meanwhile, observe in Fig. 4b, that there is an increase in the PSD in the 12–14 Hz frequency (labeled E), which significantly varies during the oscillation as compared to Fig. 4a.

Finally, observing Fig. 4b, the PSD during the event is characterized by a prominent, narrow band spectral peak(s) in the 14–15 Hz range (labeled F), which is identified to be the sustained oscillation of interest.

In addition, that for the same frequency range in Fig. 4a, that is prior to the event, there is a substantial reduction in the PSD.

Note that the spectrum peak corresponding to the oscillation (Fig. 4b, labeled F) does not appear to be sharp, e.g., like an undamped sinusoid, but rather appears to have a flat 'cupola' with two distinct peaks. This could be either the result of two closely spaced oscillatory components or a single time-varying/nonstationary oscillation. To determine what kind of oscillation it is, the time-frequency plot in Fig. 5 is obtained using Welch's method with a window length of 5 min. The oscillation of interest appears as a bright horizontal streak with its fundamental frequency gradually declining from 14.8 Hz to 14.6 Hz, which explains the flatness of average spectral estimate in Fig. 4b. This nonstationary behavior is correlated with the gradually recovering operating voltage as shown by the gray trace in Fig. 2. In Fig. 5, one can also see a peculiar spread of energy from the oscillation frequency down to 13 Hz, which corresponds to the region labeled E in Fig. 4b. While it is

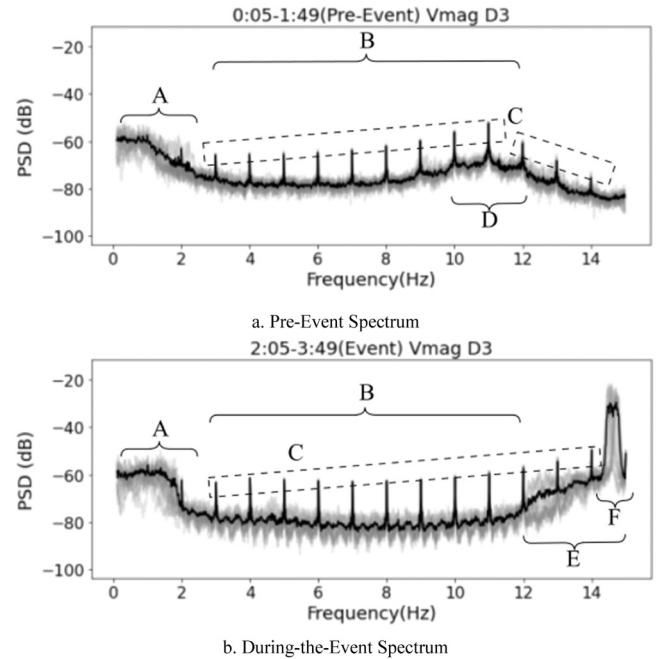


Fig. 4. Sub D3 Voltage Magnitude Spectrum Pre-and-During the Event.

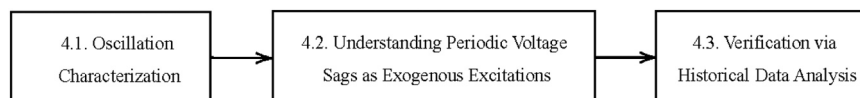


Fig. 3. Analysis Process Diagram.

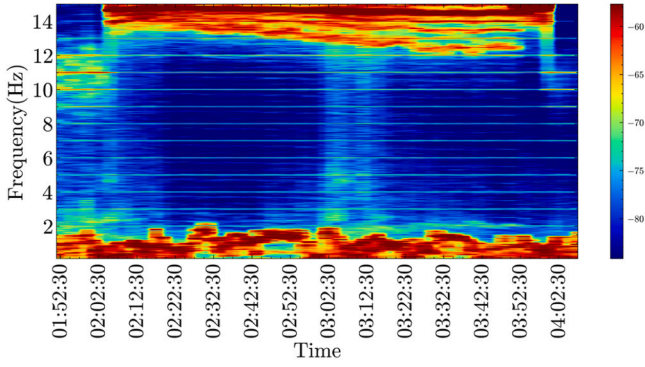


Fig. 5. Spectrogram Substation D3 Voltage During Event.

difficult to explain such response without a model, it is not uncommon in practice in nonlinear systems [27] to see such cross-frequency couplings.

Next, we further study this frequency range by analyzing the mode shape of reactive and active power flows within to make a more granular localization and to determine its governing control loop, i.e., whether it's related to a controller that regulates active or reactive power.

4.1.2. Validation against waveform (Point-on-Wave) measurement data

Because the PMU data used for the analysis above was down sampled to 30 Hz (from original 60 Hz) by the storage software, there is a risk of aliasing [2], which makes the higher frequencies (15–30 Hz in this case) indistinguishable from the base band (0–15 Hz). In our case, this would result in the 14.7 Hz oscillation being at 15.3 Hz. To determine if this is the case, it is necessary to analyze waveform data (point-on-wave data) from the oscillation period, which is typically not stored. Luckily in this case, the DFR at D3 triggered and stored a 6 min window of data around the oscillation inception, which is plotted in Fig. 6. Here, it is possible to observe that the amplitude of the waveform increases slightly in the last 1 minute of the window (approximately), which coincides with the beginning of oscillation, which we will analyze next. To compare this data with our previous observations in PMU data, we need to transform it. Note that the 60 Hz component in waveform data acts as a carrier signal with the oscillation modulated into its amplitude and/or phase, which need to be extracted for analysis. For this, we use Hilbert transform [28] of the signal from the last 1 min window containing the oscillation and plot its spectral estimate in Fig. 7. Here we can see that the oscillation component is indeed at 14.7 Hz and not displaced due to aliasing.

4.1.3. Mode shape analysis for the 14.7–14.8 Hz frequency

By analyzing the relative observability of the 14.7–14.8 Hz oscillation across the neighboring substations shown in Fig. 1, we aim to localize the source more granularly by applying the approach in [29]. This will be carried out by concentrating on a 30-min. window between 2:10 and 2:40 AM, where the oscillation is ~ 14.7 Hz. From the pre-event spectral estimates in Fig. 4, there is no prominent mode in 14–15 Hz

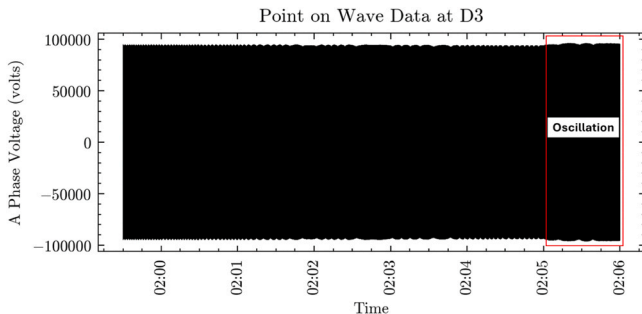


Fig. 6. Waveform Data Phase A Voltage at D3.

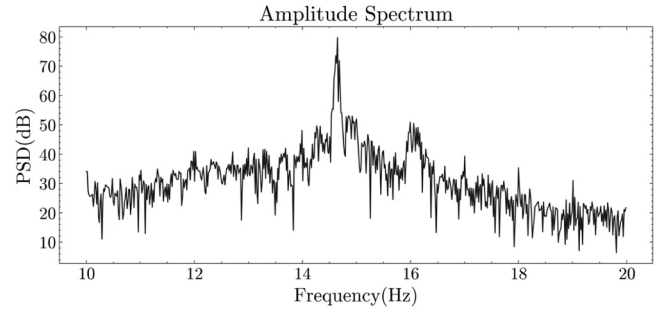


Fig. 7. Amplitude Spectrum of Phase A Voltage at D3.

range and therefore, there is no risk of resonance, which can otherwise create difficulties when identifying the true oscillation source. Consequently, this makes the source location straightforward using mode shapes. Here we use Frequency Domain Decomposition (FDD)[18] to estimate mode shapes. Given the fact that the oscillations are prominent in the measured voltage magnitude, we start by estimating the mode shape at ~ 14.7 Hz as shown in Fig. 8, where it is clearly evident that the substations in the load region (D1 and D3) participate in that oscillation to a greater extent than the other substations, hinting at a likely source. The characteristic to notice is that the mode shape of all substations are close in phase (except for C), ruling out any potential control interaction between the various devices at the different substations. Furthermore, substation C is the shortest path (electrically) to higher kV levels from substations D1 and D3, and therefore, it is considerably better regulated and has the lowest participation. To investigate this more thoroughly, we examine the mode shapes from power flows.

Fig. 9 presents the reactive power (Q) mode shape. When analyzing Volt-VAR oscillations, a source will typically be characterized by $Q-V$ oscillating nearly in phase and having the largest participation. On the other hand, a sink (successfully regulating voltage) has $Q-V$ oscillating out of phase. This can be examined by observing the mode shapes in Fig. 9, where the biggest Q mode shape magnitudes are observed on lines D1 to C and D3 to K. Furthermore, these are nearly in phase with each other, i.e. constructively contributing/absorbing the oscillation. To determine if the observed behavior is that of a source or a sink, we estimate the mode shape using total Q and P injection in data center region (D1 to C + D3 to K) and voltage magnitude in Fig. 10. It can be observed that Q and V in the data center region oscillate nearly in phase, pointing towards it being the most likely source. Although measurements are unavailable at D2, it is possible to observe that there is only a small difference between the Q oscillation entering and leaving D2; consequently, for all intents and purposes, it plays little to no role in this event. Finally, note that the two solar PV plants (at C and H) do not contribute to the oscillation, which is expected given that this oscillation

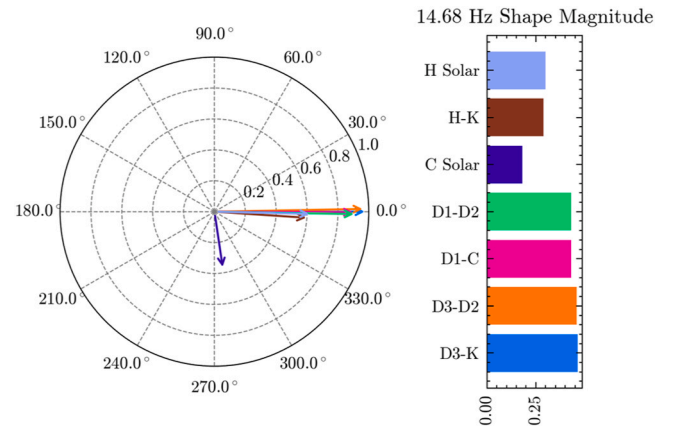


Fig. 8. Voltage Magnitude Mode Shape.

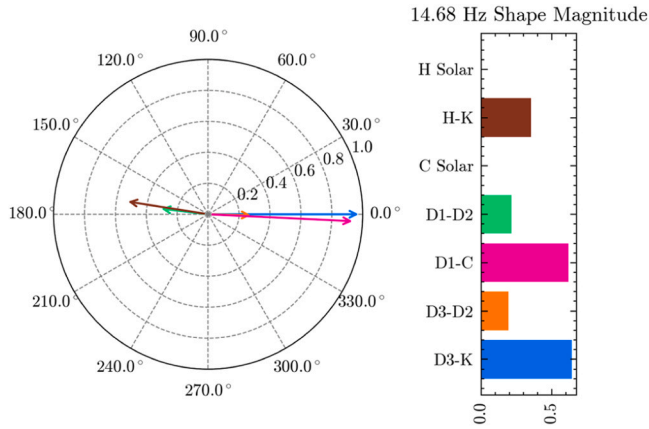


Fig. 9. Q Flow Relative Mode Shape (left) and Mode Shape Magnitude (right).

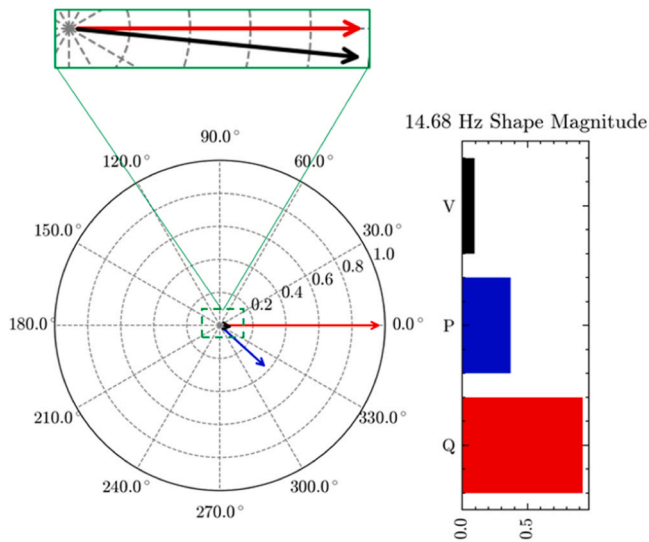


Fig. 10. V, P and Q Relative Mode Shapes (left) and Mode Shape Magnitude (right) at the Data Center Region.

occurred at night (12 AM local time) and these plants do not have capabilities to provide VARs at night.

One final aspect to investigate is to determine the mechanism that gives rise to the oscillation, i.e. whether is driven by dynamics involving the active power (P type) or reactive power (Q type). Note that there is no general rule that applies to all scenarios and is generally dependent on dynamic characteristics of the individual devices (and their controllers) participating in the oscillation. For example, an oscillation emerging from an excitation control system (Volt-VAR type) can also present an oscillation in P, and thus, it can be mistaken for P type oscillation. This is commonly seen in synchronous generators. Luckily, the study system has no such local units (hydro unit K was taken offline). Furthermore, it is mainly an IBR rich region (data centers and solar), which typically present P-Q decoupling in their controls (unless becoming unstable). In other words, what starts off as P or Q oscillation at the source will retain its character when encountering local devices. Thus, we look at the relative participation of Q and P in the data center bus (likely source) in Fig. 10. The mode shapes show that the oscillation has a relatively larger participation in Q as compared to P, pointing to a voltage control / volt-VAR problem.

4.2. Understanding periodic voltage sags as controlled inputs

Next, we study the dynamics appearing as spectral peaks at 1 Hz and

its multiples (labelled C in Fig. 4). The closest known signal type that could result in such a spectrum is a pulse train. In our previous work [30], we observed a similar phenomenon at 0.016 Hz and its multiples resulting from a voltage sag that occurs once a minute, and can be observed throughout the power network. However, in the present case, the fundamental frequency is 1 Hz and, therefore, it should be happening once a second. To confirm this, we plot the voltage magnitude at D1 in Fig. 11. One can see a sharp drop in voltage every second (indicated within the dashed box labelled A), which represents periodic voltage sags. Such sags are visible for more than a minute in Fig. 11, as indicated by the scale labelled B. Furthermore, the size of the drop increases significantly for roughly 10 sec. every minute, labelled C in Fig. 11. This could be due to the duty cycle of an data center. Although such exogenous excitation is undesirable, it can help analyse the response of the system, which is otherwise difficult to solely estimate from ambient conditions.

We take a closer look at the dynamics in the 10–12 Hz range (labelled C in Fig. 4), using the time-frequency methods summarized in Section 3. Recall that these dynamics vanish during the oscillation event and, therefore, we need to study the periods where the oscillation is not present. Here is where the exogenous excitation from the periodic voltage sags can be exploited. Fig. 12 shows a time-frequency plot of voltage magnitude at sub D1 in the 5–15 Hz range. One can see that the relevant dynamics are concentrated in 10–11 Hz as confirmed by the spectrum plot in Fig. 4. However, what cannot be observed in Fig. 4 is the temporal information that Fig. 12 provides, which shows that the mode is only observable for a short duration of ~ 10 s, separated by precisely a minute. These periods interestingly align those when voltage sags become large. Zooming into one of these ~ 10 s periods in Fig. 13 confirms this, where it is seen that the voltage sag on top of each second triggers a ringdown response. This is what we were alluding to earlier where exogenous excitations expose otherwise difficult to observe underlying system behaviour. Now, the question arises as to why this mode vanishes when the forced oscillation is happening and whether it has a relationship to it, which is addressed next through historical data analysis.

4.3. Verification via historical data analysis

Given the fact that data from similar events was found, this provided a unique opportunity to verify the conclusions made regarding the oscillation's inception and emergence. Next, we explain how such data was found, how historical data analysis was used to provided further insight into the incipient behavior prior to the emergence of the oscillation and to verify the previously made observations.

4.3.1. Detecting similar event instances and identifying common patterns

According to the analysis so far, the oscillation is concentrated between 13 and 15 Hz. Therefore, high PSD values in this frequency band characterize key periods in the historical data, which can be utilized as a

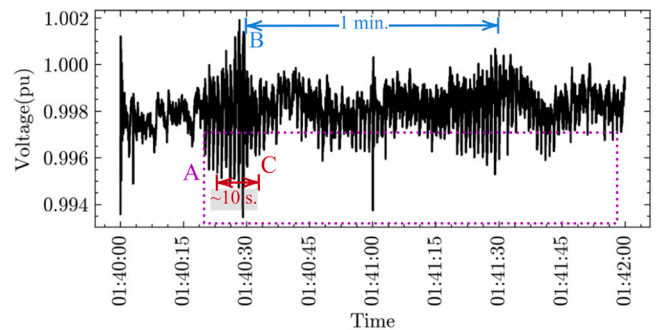


Fig. 11. Voltage Magnitude at D1 Showing a Periodic Voltage Sag Every Second.

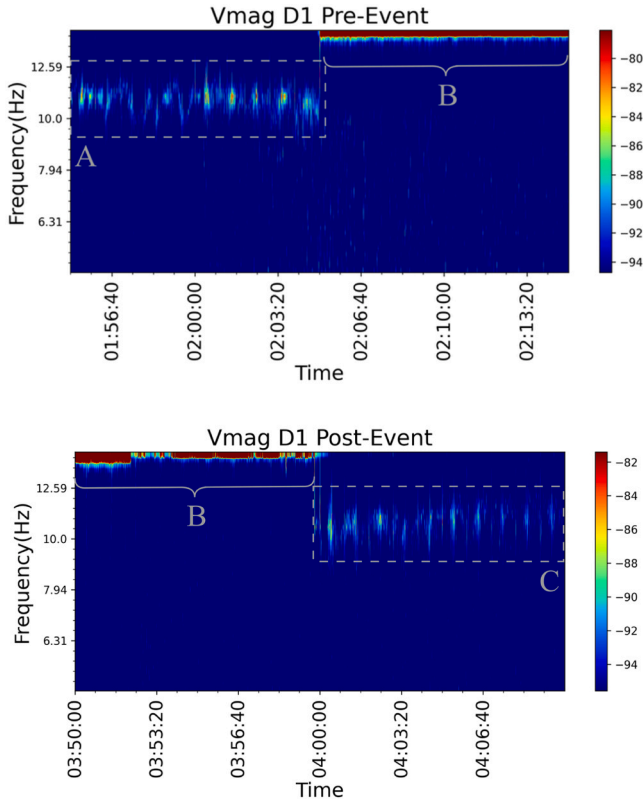


Fig. 12. 10–11 Hz Time-Frequency Representation. A: indicates energy content at ~ 10.5 Hz before the event; B: indicates the oscillation event period; C: indicates energy content at ~ 10.5 Hz after the event.

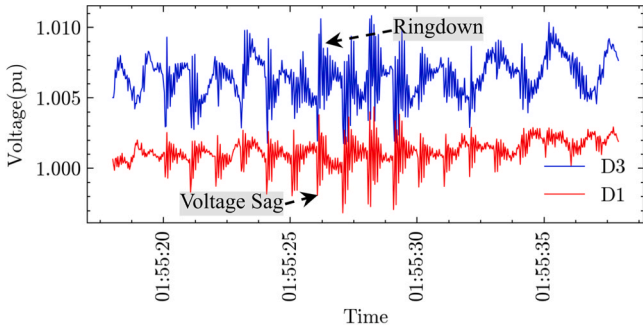


Fig. 13. Voltage Sag Exciting 10–11 Hz Dynamics.

metric to detect them, similar to what is done in RMS energy detectors [31]. Let us denote this metric by

$$S_{event}(t) = \max(\{S(t,f) \forall f \in [13, 15]\}), \quad (5)$$

where $S(t,f)$ is the PSD at time t as defined before. During system transients that are a result of faults, large load changes, etc., the PSD will have a broadband frequency domain characteristic [32]. This means that such transients also cause a significant increase in the PSD value in this range of interest (along with all other frequencies), which could lead to false positives when attempting to detect similar events. To address this issue, it is necessary to single out and filter the periods when this occurs, for which we use

$$S_{transient}(t) = \text{median}(\{S(t,f) \forall f \in [5, 15]\}). \quad (6)$$

Meanwhile, periods when the oscillation appears are characterized by

$$S_{event}(t) \gg S_{transient}(t). \quad (7)$$

Next, considering ambient conditions and in the absence of any other dynamic behavior in 13–15 Hz range, we have that

$$S_{event}(t) \approx S_{transient}(t). \quad (8)$$

as can be seen in the in Fig. 4a. Finally, since we are working with an estimate of $S(t,f)$ denoted by $\hat{S}(t,f)$, taking a log transform is advantageous in that it makes the variance of $\hat{S}(t,f)$ approximately stationary [17], thereby allowing to work across t and f , leading to the following candidate for the detector,

$$d(t) = \log(\hat{S}_{event}(t)) - \log(\hat{S}_{transient}(t)) > \text{threshold}. \quad (9)$$

Using this detection on the available historical data, the plot in Fig. 14 was obtained. In the figure, there are two major groups of operating conditions labelled A and B, respectively. Here, the distance from the black line ($y = x$) represents the metric d . It was discovered that Cluster A belongs to windows with a considerable amount of missing data and therefore a low value of signal power. Cluster B can be further divided into two clusters, depending on the distance from the black line. Here, Grubbs Test [28] can be applied on $d(t)$ in Cluster B with $\alpha = 0.05$ to detect oscillation events as a form of outliers (red dots labelled C). Note that the Grubbs test stipulates that the points must be i.i.d. normally distributed, however, this is not always the case because the grid's operational conditions affect $S(t,f)$ and vary continuously in time t , therefore having a dependence. Regardless, gross outliers are still identifiable it can clearly by observed in Cluster C shown in Fig. 14.

Next, we take a closer look at a few of the events identified (shown as red dots in Fig. 14) using the methods described in Section 3. Fig. 15 shows a 2 min window of 7 of the identified events in the time domain (left), as well as the time-frequency plane (right) obtained using the synchrosqueezed CWT. Here, the oscillation event under study is at the top, which is what we have been concentrating on so far. The previously examined 10–11 Hz dynamics which denote the inception of the ~ 14 Hz oscillation over different time periods characterizes all of these. In addition, oscillations frequently emerge and disappear without the system undergoing any noticeable changes, such as line outages, faults, etc. Note that the short-lived oscillations, which contain vital information on the underlying inception mechanism behind the oscillation, are relatively difficult to capture due to the time-frequency resolution trade-off.

4.3.2. Inferring the oscillation mechanism

The dynamics between 10 and 11 Hz, which frequently lead to the ~ 14.7 Hz oscillation, are carefully examined next. As was previously observed, voltage sags occur precisely once every second for about 10 s every minute allow us to examine the system response. For this analysis, we select the window in each of the previously discussed cases (see Fig. 15) before the oscillation event. The analysis is carried out as follows:

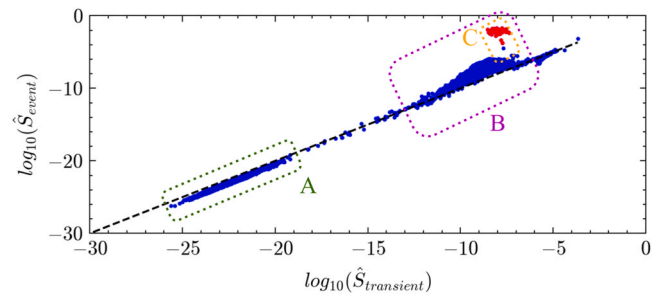


Fig. 14. Historical Spectral Estimates. A: cluster with significant missing data; B: cluster with potential events; C: events identified as a form of outlier using Grubbs Test.

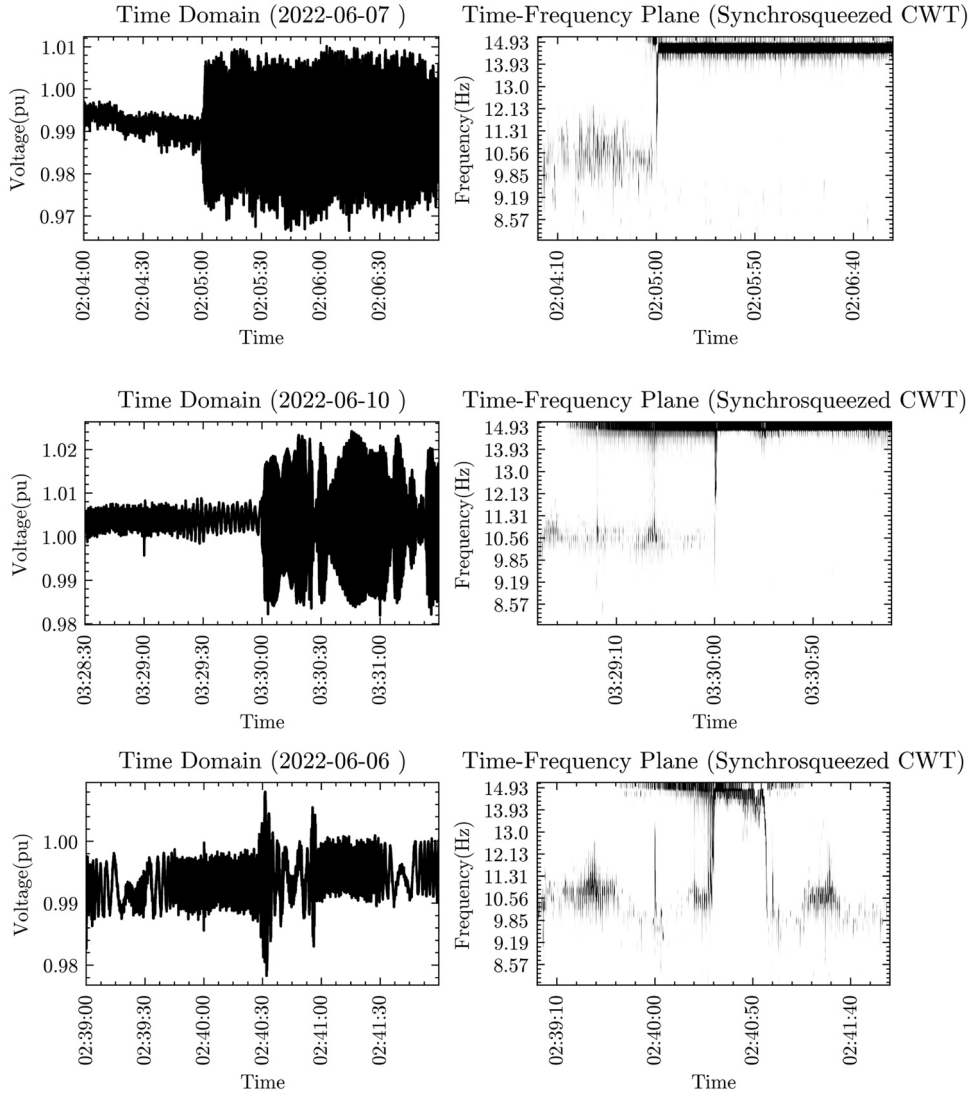


Fig. 15. Identified Oscillations, Time Domain (Left), Time-Frequency Plane (Right).

- First, we divide the window into 10 segments highlighted grey in Fig. 16 (left column plots).
- Next, for each segment, we use voltage magnitudes at substations D1 and D3, detrend, and band pass filter them to 9–15 Hz and fit a second-order model to estimate the dominant mode using Dynamic Mode Decomposition [33].
- Finally, the trajectory of the estimated mode is plotted in the damping-frequency plane in Fig. 16. As can be observed, a critical mode in the 10- to 11-Hz range approaches dangerously low damping in all cases, starting as a small signal stability problem followed by a sudden increase in frequency when settling on a limit cycle (which are the oscillation events). It is worth noting that the nonlinear system response makes the linear and low-order model employed to estimate the damping of this mode unsuitable for analysis at unstable periods (when the damping < 0), and as a result, the mode estimates obtained under such conditions are unreliable.

Thus, we can see that a 10–11 Hz mode local to data center becomes unstable with changing operating conditions, which is the inception of the oscillation at 14.7 Hz. Observe that this oscillation is different from forced oscillations [34], which are characterized by an external oscillatory input usually due to an equipment malfunction. Since state

variables are not measured, explaining why an instability resulted in an oscillation will require deductive reasoning using observations in output measurements. Following an equilibrium becoming unstable, the system trajectory can do one of two things:

- **Go unbounded** – In all of the oscillation events in Fig. 15, we can see that once the original equilibrium re-stabilizes due to organically changing operating conditions, which took anywhere from a few seconds to hours, the trajectory immediately returns to an equilibrium marked by the oscillation vanishing abruptly. For this to happen, the trajectory must be bounded within the stability region [35] of the re-stabilized equilibrium, in some cases for as long as 2 Hrs, which makes the unbounded case highly unlikely.
- **Enter some invariant set** other than the original equilibrium (limit cycle, equilibrium, etc). In this case, the trajectory remains bounded, which makes it a more likely scenario in our case. Observe that the trajectory does not settle to a new equilibrium point, as the output measurement changing continuously (or rather, it oscillates). Given that the response behaves as a periodic motion, a limit cycle [36] (which is an isolated periodic orbit), the most likely scenario is that what we observe is in fact a limit cycle.

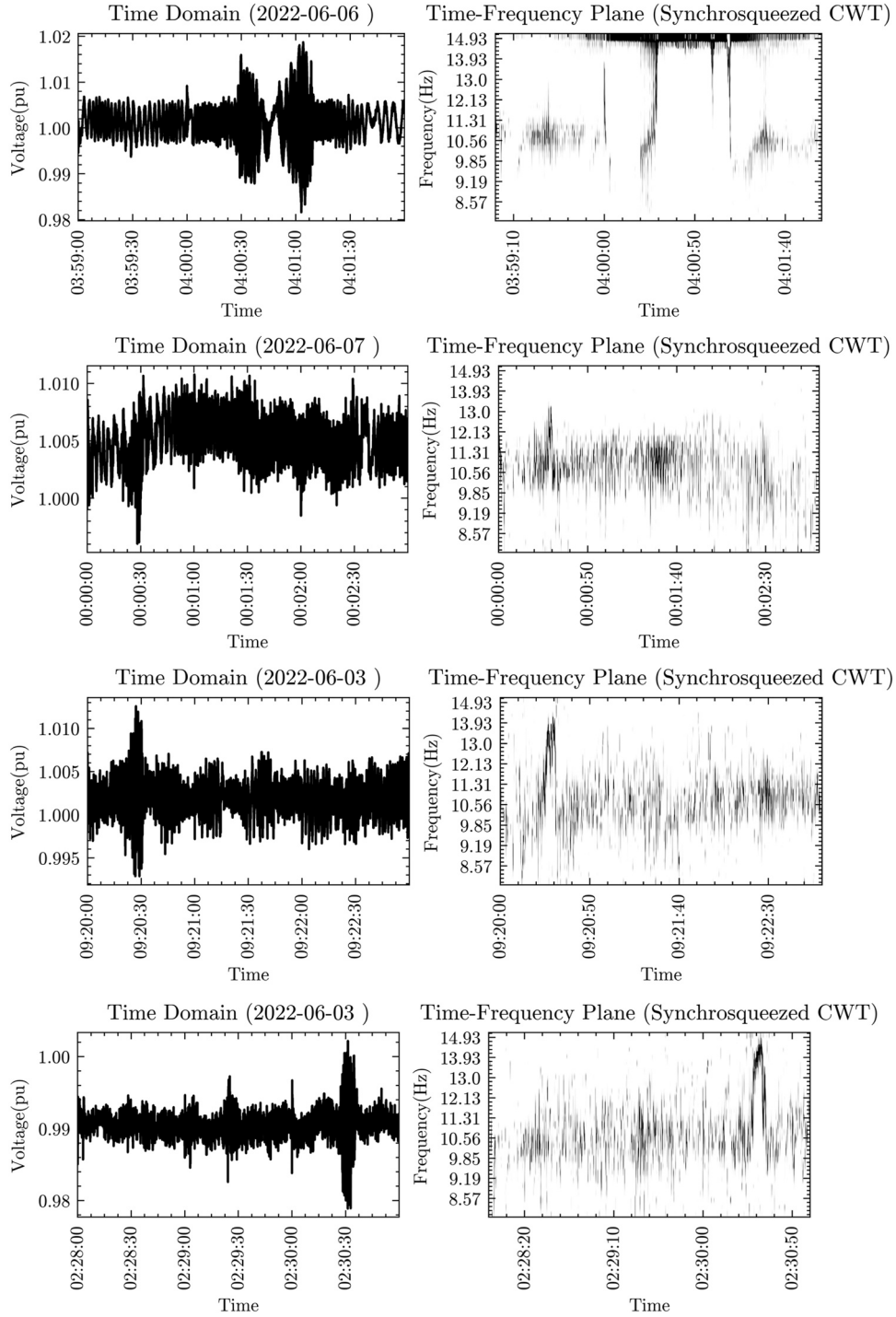


Fig. 15. (continued).

Proceeding with the limit cycle hypothesis, we would like to understand the mechanism that led to its creation. Noting that the creation of limit cycle was preceded by a pair of complex eigen values (oscillatory mode) crossing the imaginary axis, the initial assumption is that we are observing a supercritical Hopf bifurcation. Such kind of bifurcation involves a creation of stable limit cycle at the instance of a stable equilibrium destabilizing, thereby catching the system trajectory. Note that a distinguishing characteristic of this mechanism is that the limit cycle

created starts off with essentially a zero radius and grows as the system conditions change (for the worse). However, in all the events analyzed, as soon as the system becomes unstable, the oscillation starts off at a large radius even though the operating point is changing gradually (particularly in the June 6th events), making the validity of this hypothesis questionable. A more common mechanism behind creation of limit cycles from controller instability, stems from the fact that most real world controllers (particularly those in power electronic-based devices)

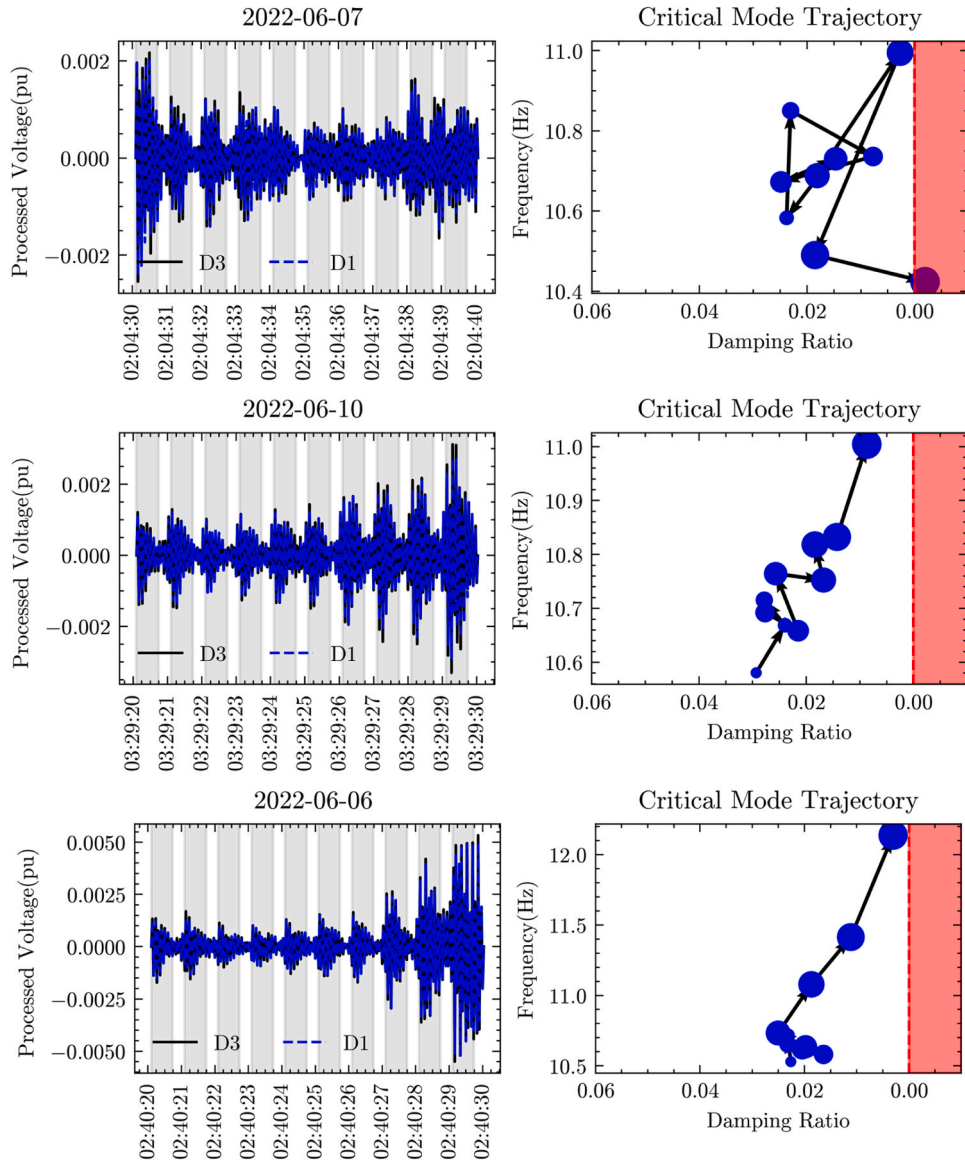


Fig. 16. Evolution of the Damping of the 10–11 Hz Dynamics.

are equipped with hard limiters (such as under/over excitation limits in voltage controllers) to protect the equipment. [37] and [38] presented a mechanism where the hard limits chop the state space in a way that creates a cyclic motion (i.e., a limit cycle), once the trajectory goes unstable and hits those limits. These limits are usually away from the original normal operating point (equilibrium), thereby resulting in a relatively large amplitude oscillation, governed by the limits. In other words, the trajectory grows and enters a cyclic motion, which seems to better explain our oscillations stemming from an unstable UPS in the data center. To demonstrate this concept in simulation, we have created a simple proof of concept simulation of a control system with limiters, which is described in the Appendix.

4.3.3. Long term analysis of critical mode (10–11 Hz)

Finally, the long-term behaviour of the critical mode of 10–11 Hz

critical mode, captured by

$$S_{mode}(t) = \max(\{S(t,f) \forall f \in [10, 11]\}) \quad (10)$$

and $S_{transient}(t)$ (see Equation (11)), is shown in Fig. 17. Periods with a disproportionately high value are labelled A. These are consistent with a local transmission line outage in the vicinity of the data center, which reduces the damping of the critical mode and raises the PSD value (and hence, $S_{mode}(t)$), as explained above. The significant increase that begins in May 2022 (labelled B) and continues until the day before the oscillation events in June is important to highlight. This suggests that, in terms of data center performance, the grid may be operated regularly under more stressed conditions, that is, with a decrease in stability margin. This is in line with our analysis in Section 4.1, and further points to the data center region as the culprit.

Based on the analysis above, the measurements in this data center

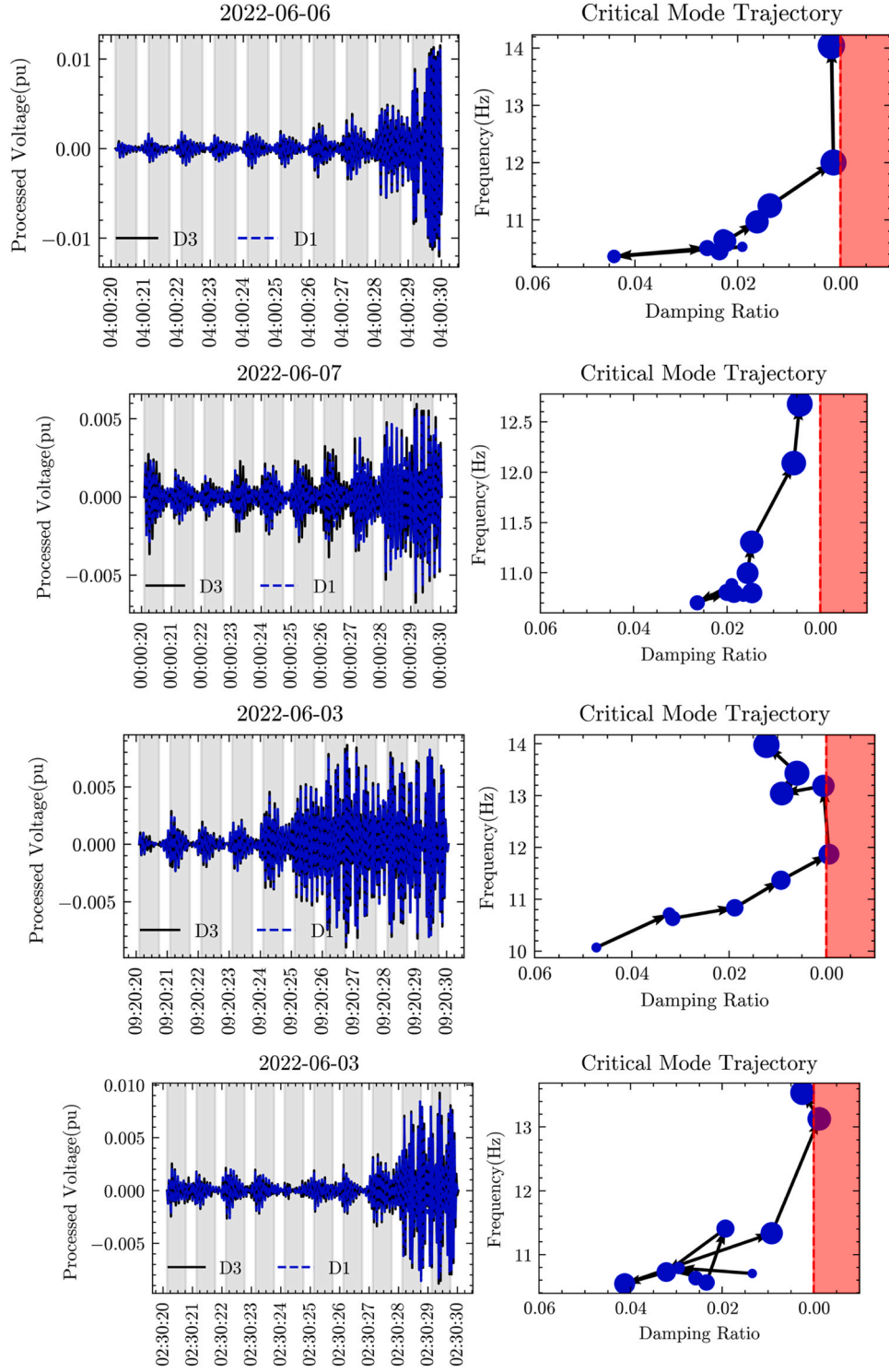


Fig. 16. (continued).

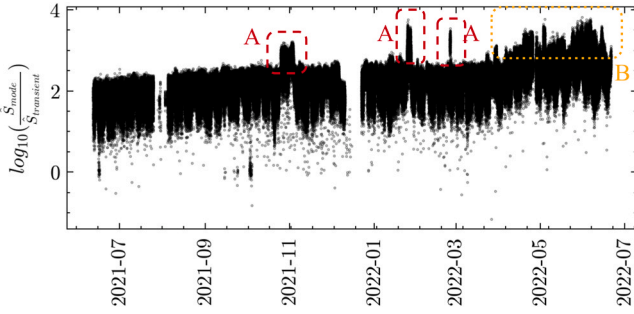


Fig. 17. 10–11 Hz Signal Power of the Voltage Magnitude at D3. A: Periods with disproportionately high value in $S_{mode}(t)$. B: Significant increase in $S_{mode}(t)$.

region in ambient conditions can be modeled as the response of a linear system (with 10–11 Hz critical mode) driven by a periodic forcing applied every second. Once unstable, a limit cycle > 14 Hz is formed. While it is not possible in real time to “infer” what exactly is happening, at the minimum, it should be possible to track the known dynamics as follows:

1. Because the periodic forcing will result in the system’s response to be characterized by a transient every second, it is possible to detect its occurrence similarly to how faults or voltage sags are detected, for example using wavelets [39].
2. Having detected each of the aforementioned transients, data can be collected for each < 1 second window and the critical mode (frequency and damping) can be estimated using any parametric approach (Prony, ARMA, etc.) [33], similarly to what is done in 4.3.2.
3. Detecting whether the mode has become unstable is fairly easy in this case, since it results in a sustained oscillation, for which there are many techniques available [31]. These operate by detecting an increase in power in the corresponding frequency band (14–15 Hz in this case).

5. Conclusion

This paper investigated a real-world 14.7 Hz oscillation in the Dominion Energy service area that originates from a data center-rich region using measurement-based analysis. The root cause is attributed to the instability of a 10–11 Hz mode related to the data center’s UPS system interacting with specific grid conditions, which manifests as the ~ 14.7 Hz oscillation (limit cycle) upon destabilization. A summary of the main findings of our analysis is as follows:

- Historical data analysis of similar oscillation events revealed intermittently observed 10–11 Hz dynamics.
- The aforementioned intermittency was caused by an unrelated periodic forcing in the system, which manifested as voltage sags every second. This force was responsible for sufficiently stimulating the 10–11 Hz mode, which would otherwise be difficult to determine from ambient data. Tracking its damping indicated that the oscillations were always preceded by a loss of stability for this mode.
- Due to a lack of models, a detailed investigation of measurement data combined with deductive reasoning led to the conclusion that the observed ~ 14.7 Hz oscillation was a limit cycle caused by an interaction between an unstable controller and limiter.
- Finally, in this work, it was crucial for the data center owner to verify the analysis results, as it is not owned by the utility. The owner of the asset confirmed that the oscillations were caused by the input stage

of the UPS units that are deployed at the data center, which have different ancillary service capabilities [4] [5].

The correct characterization of this complex dynamic process resulted in minimizing the liability on its emergence for the utility, a benefit that will have lasting effects in addressing growing stability issues at the transmission level due to the increase of converter-interfaced loads and energy sources.

It is necessary to emphasize the limits of a purely data-driven approach to understanding the mechanism behind real-world oscillations:

- *Need for physics-based models:* Without access to the proprietary controller and power stage models for the data center, it is impossible to pinpoint the device characteristics or components causing the critical dynamics. More significantly, measurement data cannot replace models for what-if analysis or design of countermeasures required to avert similar problems in the future.
- *Masking dynamics:* Deadbands in real-world controllers can obscure the controller response in ambient data, making it difficult to detect emerging stability issues.
- *Effects of ambient process on estimation:* Because of the stochastic character of the power system’s response in ambient settings, we deal with estimates rather than deterministic quantities. As a result, care must be taken not to over-interpret results. For example, in Fig. 16, it was necessary to filter the data using a high pass filter for frequencies > 9 Hz range before fitting a low order time series model to estimate the critical mode accurately (a low order is favored as only 1–2 modes need to be estimated). Fitting a time series model without prior knowledge of what to target (e.g., the adequate model order) is challenging to do in practice [40], resulting in misleading, spurious dynamics that have no physical meaning and may lead to misinterpretations or even false results [41].
- *Limitations of synchrophasor data:* It is always important to verify results against point on wave data [2]. Synchrophasor devices often prefilter around 60 Hz, which can heavily distort relatively fast dynamics. Furthermore, inter-harmonics near 60 Hz might emerge and be misinterpreted as oscillations in synchrophasor data.
- *Mistaking all periodic signal components for natural or forced oscillations:* Real-world systems present many bounded, periodic inputs, which are frequently caused by load duty cycles [42]. In particular, as described in this work, the periodic voltage sags (see Fig. 13) may have been misunderstood as forced oscillations, despite the fact that the analysis using long-term data reveals them to be tightly limited and not necessarily caused by equipment malfunction or instability.
- *Time-frequency resolution trade-off:* It is impossible to perfectly localize dynamics in time-frequency plane, as explained by uncertainty principles, which makes analysis of fast changing dynamics difficult. As an example, in the spectral estimate in Fig. 4, the 0–2 Hz region labelled A appears like a well damped/stable behavior (typically characterized by a flat peak in spectrum). However, it represents a sustained, low magnitude oscillation (0 damping) whose frequency varies rapidly in 0–2 Hz range and thus, its power is distributed almost uniformly.
- *Data quality:* Data dropouts for extended periods are common in real-world data, which can make analysis difficult. For example, in Fig. 14, when trying to detect oscillation events in past data, the missing data cluster labelled A results in relatively small spectral estimate values, can potentially make the energy detector in (9) to give a false negative (trigger off).

CRedit authorship contribution statement

Jones Kevin: Resources, Project administration, Investigation.
Purcell Thomas: Data curation. **Delaree Jr. Jaime:** Writing – review & editing, Resources, Project administration. **Vanfretti Luigi:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Conceptualization. **Mishra Chetan:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software,

Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix. – Proof of Concept Simulation

In this section we attempt to replicate the phenomena observed and analyzed in the paper through the simulation of a simple linear control system shown in Fig. 18. This approach is helpful to understand the underlying mechanism, given that what we saw in the actual system response is a small signal stability issue characterized by:

1. Voltage Sags, i.e., an impulse type input once a second, and a
2. Limit Cycle through an interaction between unstable controller and limiter.

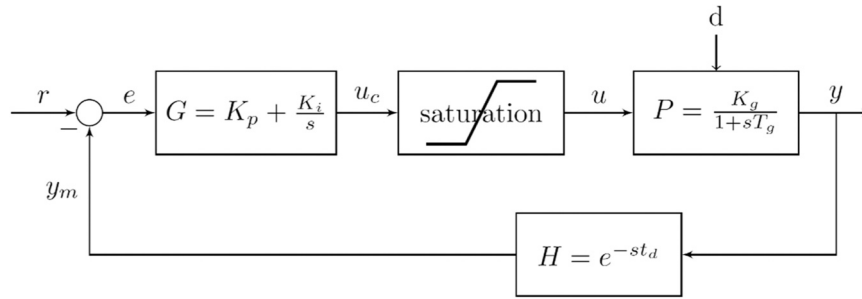


Fig. 18. Test System for Proof-of-Concept Simulation

In Fig. 18, the PI controller $G(s)$ regulates y (read voltage) to a reference value r . The control output u_c (read reactive power Q) passes through a hard limiter, with output u of the form $u = \begin{cases} u_{\min}, u_c < u_{\min} \\ u_c, u_{\min} \leq u_c \leq u_{\max} \\ u_{\max}, u_c > u_{\max} \end{cases}$. The system (plant) to be controlled is modeled by a first order filter with gain K_g and time constant T_g , with T_g set to a small number to mimic the fast response of an inverter-based plant connected to the grid. This is typical of subsystems with no other dynamical devices in close vicinity, with external consequently behaving like a Thévenin equivalent. y is also impacted by external disturbance d , in the form of periodic sags to mimic the input applied to the actual system. The controller has a feedback delay of t_d seconds, modeled by the H block.

Small Signal Stability Analysis

Setting arbitrary values for control parameters $K_p = 1$, $K_i = 2$, grid parameters $T_g = 0.1$ and feedback delay $t_d = 0.1$, setting $r = 0$, we plot the poles of the closed loop transfer function between ambient excitation d and output y in Fig. 18 by varying grid gain K_g in $[0.1, 3]$ range. Note that here, the saturation block plays no role, as we are analyzing small signal stability. Here we can see that the critical value of $K_g = 2.2$, where the lowest frequency mode ~ 3.1 Hz becomes unstable (moves from left to right and becomes negatively damped). From here onwards, we will refer to it as the critical mode.

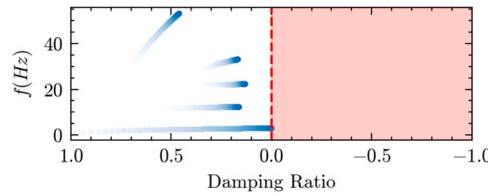


Fig. 19. Dominant Modes for the Proof-of-Concept Simulation Model

Setting a stable value of $K_g = 1.5$ and making the applying the periodic sags at 1 second interval together with Gaussian white noise to d , the model is simulated. Observe in Fig. 19 that this signal excites and makes the critical mode ~ 3 Hz observable. This can be confirmed from y 's spectral estimate in Fig. 20, where a “hump” can be observed at ~ 3 Hz along with a collection on spikes at intervals of 1 Hz (periodic sags). Please note that since all other modes are sufficiently damped (and stable), it is not easy to excite them and consequently, to observe them.

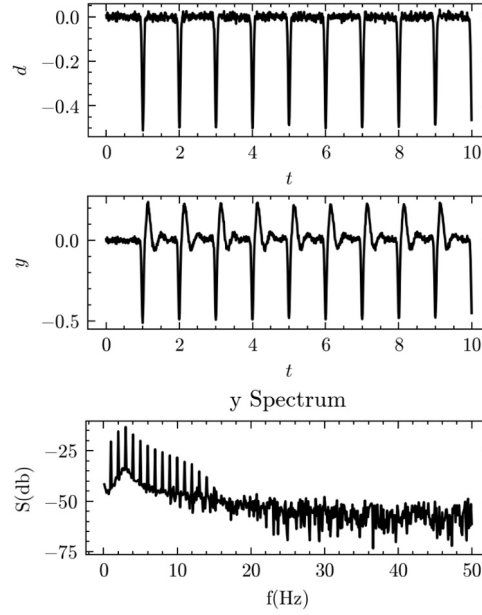


Fig. 20. Periodic Sag (d) Exciting Stable Critical Mode (y) and Computed Spectrum of the Response ($S(\text{db})$)

Unstable Mode Giving Rise to Limit Cycle and the Impact of Limiters

In this experiment, we set K_g to a marginally unstable value of 2.3 and analyze the effect of control limiters in the “saturation” block of Fig. 18 being set at $u_{\min} = -5$, $u_{\max} = +5$. Fig. 20 shows the system trajectory in u – y plane, where the difference between “No Limiter” (top left, LHS) and “Limiter” (top right, RHS) plots is that the limiter (given by horizontal blue lines on the top right) is disabled (in LHS) and enabled (in RHS). The impact of the limiter is to “chop” the phase portrait in such way that what should have been a trajectory spiraling out of control (as seen on LHS plot) due to the unstable equilibrium at $(0, 0)$, now becomes a periodic solution (limit cycle, seen as the dark black orbit in RHS). This response, which bounds the system dynamics through the limits, also allows for the system to be restabilized once the equilibrium point becomes stable.

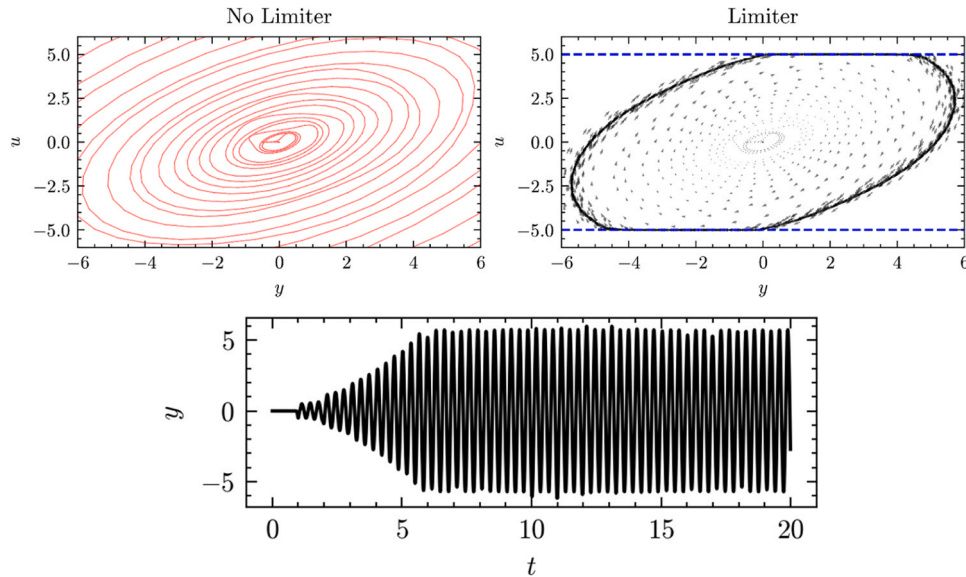


Fig. 21. Unbounded Response and Bounded Response (due to Limiter) Resulting in a Limit Cycle

Data availability

The data that has been used is confidential.

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